

1st Midterm
ME 201A: Engineering Thermodynamics

Date 21/11/2022

Exam Duration: One Hour

Maximum Marks: 10

1. A body of constant heat capacity c_p and initial temperature T_1 is placed in contact with a heat reservoir at temperature T_2 and comes to thermal equilibrium with it. If $T_2 > T_1$, calculate the entropy change of the universe and show that it is always positive. **[2]**
2. Derive the expression of availability or maximum work done by the closed system in the change of state from 1 to 2 by interacting only with the surroundings at p_o, T_o . **[5]**
3. Derive the vander Waals constants in terms of critical properties. **[3]**

“MATERIAL TECHNOLOGY” I MID-TERM EXAM 2022-23

MM: 20

Date: 21/11/2022

1. Sketch and explain all type of crystal structures. (5)
2. Define the term: - number of atoms in unit cell, coordination number and atomic packing factor with examples. (5)
3. Draw a neat sketch of Fe_3C – C diagram labeling all important phases and constituents. (5)
4. Explain and sketch different dislocations. (5)

MBM University, Jodhpur
Department of Mechanical Engineering

B.E. IInd Year IIIrd Semester

First Mid-Term Examination Session 2022-23

ME-203A: Kinematics of Machines

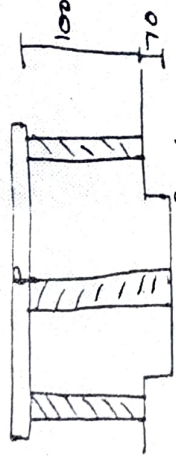
Time: 1 hour

Maximum Marks – 20

- Q. 1. (a) Discuss the effect of slip and creep of belt on the pulleys on the velocity ratio of a belt drive. (3)
- (b) Derive the relation for the ratio of belt tensions in a flat-belt drive. (3)
- (c) A casting having a mass of 100 kg is suspended freely from a rope. The rope makes 2 turns round a drum of 300 mm diameter rotating at 24 rpm. The other end of the rope is pulled by a man. Calculate the force required by a man, power to raise the casting. Take $\mu = 0.3$. (4)
- Q. 2. (a) Sketch and explain the various inversion of four-bar chain mechanism. (4)
- (b) Explain different kinds of kinematic pairs giving example for each one of them. (6)

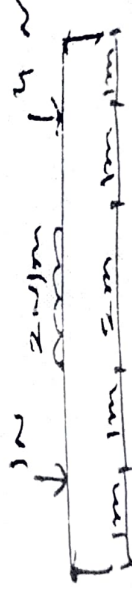
TIME: 60 MIN MM:20

1. Two brass rods and one steel rod together support a load as in fig.. If the stresses in bars and steel are not to exceed 60 N/mm^2 and 120 N/mm^2 . find the safe load that can be supported. $E_{\text{Steel}} = 2 \times 10^5 \text{ N/mm}^2$ $E_{\text{bars}} = 1 \times 10^5 \text{ N/mm}^2$, Area Steel = 1500 mm^2 , Area brass = 1000 mm^2



(6)

2. Draw the SFD and BMD for the following:



(8)

3. Derive bending equation and state the assumptions made.

OR

4. A timber joist of 6 meters span has to carry a load of 500 kg/m . Find the dimensions of joist, if the maximum permissible stress is limited to 80 Kg/cm^2 . The depth of joist has to be twice the width. (6)

Note: Attempt any two questions.

Q (1) Given the following data

x:	10°	20°	30°	40°	50°	60°	70°	80°
y:	0.9848	0.9397	0.8660	0.7660	0.6428	0.5	0.3420	0.1737

Evaluate (i) $y(25^\circ)$ (ii) $y(73^\circ)$

Q (2) (i) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using trapezoidal rule with $h=0.2$ and hence determine the values of π

(ii) Evaluate $\int_0^{\pi/2} \sqrt{\cos \theta}$ do by dividing the interval into eight equal parts.

Q (3) Determine y at $x=21$ given the following:

x:	10	15	22.5	33.75	50.625	75.937
y:	0.3	0.675	1.519	3.417	7.689	17.3

MBM University Jodhpur

Department of mechanical engineering

Subject- MM

Max Marks-10

Time: 1 Hr

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| Q.1 Explain the importance of impedance loading and matching for a measuring instrument. | 3M |
| Q.2 Explain the working principle of Piezo-electric accelerometer and Inductive pick-up tachometer. | 4M |
| Q.3 Explain the following terms | |
| a. Hysteresis and dead zone | |
| b. Precision and repeatability | |
| c. Sensitivity and Threshold | 3M |